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CSE-10.

UNIT-I

1. $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{\sin 2x}{\log x}$

$\Rightarrow \frac{dy}{dx} + \left(\frac{1}{x\log x}\right)y = \frac{\sin 2x}{\log x}$

$I.F = e^{\int P(x)dx} = e^{\int \frac{1}{x\log x}dx}$

$= e^{\log(\log x)}$

$I.F = \log x$

$G.S \Rightarrow y(I.F) = \int Q(I.F)dx + C$

$y.\log x = \int \left(\frac{\sin 2x}{\log x} \cdot \log x\right)dx + C$

$y.\log x = -\frac{\cos 2x}{2} + C$ //

2. $\frac{dy}{dx} + 2y = e^x + x$, $y(0) = 1$

$\frac{dy}{dx} + 2y = (e^x + x)$

$I.F = e^{\int P(x)dx} = e^{\int 2dx}$

$I.F = e^{2x}$

$G.S \Rightarrow y(I.F) = \int Q(I.F)dx + C$

$y.e^{2x} = \int [(e^x + x)e^{2x}]dx + C$

$y.e^{2x} = \int [e^{3x} + x.e^{2x}]dx + C$

$y.e^{2x} = \frac{e^{3x}}{3} + \int [x.e^{2x}]dx + C$

$y.e^{2x} = \frac{e^{3x}}{3} + \frac{1}{2}x.e^{2x} - \int \left(\frac{e^{2x}}{2}\right)dx + C$

$$y \cdot e^{2x} = \frac{e^{3x}}{3} + \frac{x \cdot e^{2x}}{2} - \frac{1}{4} e^{2x} + C$$

$$y = \frac{e^{x}}{3} + \frac{x}{2} - \frac{1}{4} + C \cdot e^{-2x}$$

When $y(0) = 1$

$$1 = \frac{1}{3} + 0 - \frac{1}{4} + C$$

$$1 = \frac{4-3}{12} + C$$

$$C = 1 - \frac{1}{12}$$

$$C = 11/12$$

$$\therefore \quad y = \frac{e^{x}}{3} + \frac{x}{2} - \frac{1}{4} + \frac{11}{12} e^{-2x}$$

3) $dr + (2r\cot\theta + \sin 2\theta)\, d\theta = 0$

$$\frac{dr}{d\theta} + 2r\cot\theta + \sin 2\theta = 0$$

$$\frac{dr}{d\theta} + (2\cot\theta) r = -\sin 2\theta$$

$$I.F = e^{\int (2\cot\theta)\, d\theta} = e^{2 \cdot \log(\sin\theta)}$$

$$I.F = \sin^2\theta$$

$$G.S = r \cdot I.F = \int (Q(\theta) \cdot I.F)\, d\theta + C$$

$$r \cdot \sin^2\theta = \int (\sin 2\theta \cdot \sin^2\theta)\, d\theta + C$$

$$r \sin^2\theta = -\int (2\sin\theta\cos\theta \cdot \sin^2\theta)\, d\theta + C$$

$$r \sin^2\theta = -2\int (\sin^3\theta \cdot \cos\theta)\, d\theta + C$$

$$r \sin^2\theta = -2 \cdot \sin^4\theta / 4_2 + C$$

$$\therefore \quad r \sin^2\theta = -\frac{\sin^4\theta}{2} + C_4$$

$\sin\theta = t$

$\cos\theta\, d\theta = dt$

$t^3\, dt$

$\frac{t^4}{4}$

$(1+y^2)dx = (\tan^{-1}y - x)dy$

$$(1+y^2)\frac{dx}{dy} = \tan^{-1}y - x$$

$$\frac{dx}{dy} = \frac{\tan^{-1}y - x}{1+y^2}$$

$$\frac{dx}{dy} + \left(\frac{1}{1+y^2}\right)x = \frac{\tan^{-1}y}{1+y^2}$$

$$I.F = e^{\int P(y)dy} = e^{\int \frac{1}{1+y^2}}$$

$$I.F = e^{\tan^{-1}y}$$

$$G.S = x.e^{\tan^{-1}y} = \int \left[\frac{\tan^{-1}y}{1+y^2}.e^{\tan^{-1}y}\right]dy + c$$

let $\tan^{-1}y = t$

$$\frac{1}{1+y^2}dy = dt$$

$$x.e^t = \int (t.e^t)dt + c.$$

$$x.e^t = e^t(t-1) + c.$$

$$x.e^{\tan^{-1}y} = e^{\tan^{-1}y}(\tan^{-1}y - 1) + c.$$

$$x = \tan^{-1}y - 1 + c_1 \quad // \quad (\because c_1 = c.e^{\tan^{-1}y}).$$

5. $(2xy + y - \tan y)dx + (x^2 - x\tan^2 y + \sec^2 y)dy = 0$

Given $(2xy + y - \tan y)dx + (x^2 - x\tan^2 y + \sec^2 y)dy = 0$

It is of the form $M(x,y)dx + N(x,y)dy = 0$

here $M = 2xy + y - \tan y$

$N = x^2 - x\tan^2 y + \sec^2 y$

To check condition for exactness.

$$\frac{dM}{dy} = 2x + 1 - \sec^2 y$$

$$= 2x - \tan^2 y$$

$$\frac{dN}{dx} = 2x - \tan^2 y$$

$$\therefore \frac{dM}{dy} = \frac{dN}{dx}$$

Given equation is exact differential equation

$$G.S = \int (2xy + y - \tan y)\,dx + \int \sec^2 y\,dy = C$$

$$= x^2y + xy - x\tan y + \tan y = C.$$

6) $(y^2 e^{xy^2} + 4x^3)\,dx + (2xye^{xy^2} - 3y^2)\,dy = 0$

Given $(y^2 e^{xy^2} + 4x^3)\,dx + (2xye^{xy^2} - 3y^2)\,dy = 0$

It is of the form $M(x,y)\,dx + N(x,y)\,dy = 0$

$$M = y^2 e^{xy^2} + 4x^3$$

$$y = 2xye^{xy^2} - 3y^2$$

To check for exactness

$$\frac{dM}{dy} = 2y \cdot e^{xy^2} + 2y^3 x \cdot e^{xy^2}$$

$$\frac{dN}{dx} = 2ye^{xy^2} + 2xy \cdot e^{xy^2} \cdot y^2$$

$$= 2y \cdot xy^2 + 2y^3 x \cdot e^{xy^2}$$

$$\frac{dM}{dy} = \frac{dN}{dx}$$

Given equation is exact differential equation.

$$G.S = \int [y^2 e^{xy^2} + 4x^3]\,dx + \int (-3y^2)\,dy = C$$

$$= \frac{y^2 \cdot e^{xy^2}}{y^2} + \frac{4x^4}{4} - \frac{3y^3}{3} = C$$

$$= e^{xy^2} + x^4 - y^3 = C.$$

$$\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$$

$$\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$$

$$\frac{dy}{dx} = -\left[\frac{y\cos x + \sin y + y}{\sin x + x\cos y + x}\right]$$

$$(\sin x + x\cos y + x)\,dy = -(y\cos x + \sin y + y)\,dx$$

$$(y\cos x + \sin y + y)\,dx + (\sin x + x\cos y + x)\,dy = 0$$

$M = y\cos x + \sin y + y$

$N = \sin x + x\cos y + x$

$$\frac{dM}{dy} = \cos x + \cos y + 1$$

$$\frac{dN}{dx} = \cos x + \cos y + 1$$

$$\frac{dM}{dy} = \frac{dN}{dx}.$$

Given equation is Exact differential equation.

$$G.S = \int (y\cos x + \sin y + y)\,dx + \int 0\,dy = c$$

$$= y\sin x + x\sin y + xy = c_1$$

8) $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0$

Given $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0$ — ①

$M = x^2y - 2xy^2$

$N = -(x^3 - 3x^2y)$

$$\frac{dM}{dy} = x^2 - 4xy$$

$$\frac{dN}{dx} = -3x^2 + 6xy$$

$$\frac{dM}{dy} \neq \frac{dN}{dx}$$

It is not a Exact differential equation.

Here M, N are homogeneous functions of degree '3'

$$Mx + Ny = x^3y - 2x^2y^2 - x^3y + 3x^2y^2$$

$$= x^2y^2 \neq 0$$

$$I.F = \frac{1}{Mx+Ny} = \frac{1}{x^2y^2}$$

eqn ① × I.F

$$\rightarrow \frac{(x^2y - 2xy^2)}{x^2y^2}dx - \frac{(x^3 - 3x^2y)}{x^2y^2}dy = 0$$

$$\Rightarrow \left[\frac{1}{y} - \frac{2}{x}\right]dx - \left[\frac{x}{y^2} - \frac{3}{y}\right]dy = 0$$

This is of the form $M_1dx + N_1dy = 0$

$$M_1 = \frac{1}{y} - \frac{2}{x}$$

$$N_1 = \frac{3}{y} - \frac{x}{y^2}$$

$$\frac{dM_1}{dy} = -\frac{1}{y^2}, \quad \frac{dN_1}{dx} = -\frac{1}{y^2}$$

$$\frac{dM_1}{dy} = \frac{dN_1}{dx}$$

Given is an Exact differential equation.

$$G.S = \int \left[\frac{1}{y} - \frac{2}{x}\right]dx + \int \frac{3}{y}dy.$$

$$= \frac{x}{y} - 2\log x + 3\log y + c$$

$$= \frac{x}{y} + \log\frac{y^3}{x^2} = c_1$$

1) $y(xy\sin xy + \cos xy)\,dx + x(xy\sin xy - \cos xy)\,dy = 0$

Given $(xy^2\sin xy + y\cos xy)\,dx + (x^2y\sin xy - x\cos xy)\,dy = 0$ — (1)

It is of the form $M(x,y)\,dx + N(x,y)\,dy = 0$

$M = xy^2\sin xy + y\cos xy$

$N = x^2y\sin xy - x\cos xy$

$$\frac{\partial M}{\partial y} = x^2y^2\cos xy + 2xy\sin xy - xy\sin\cos xy + \cos xy$$

$$= x^2y^2\cos xy + xy\sin xy + \cos xy$$

$$\frac{\partial N}{\partial x} = x^2y^2\cos xy + 2xy\sin xy + xy\sin xy - \cos xy$$

$$= x^2y^2\cos xy + 3xy\sin xy - \cos xy$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}.$$

This is not Exact differential equation, homogeneous function

This is of the form $y^n f(x,y)\,dx + x^n g(x,y)\,dy = 0$

consider $Mx - Ny = x^2y^2\sin xy + xy\cos xy - x^2y^2\sin xy + xy\cos xy$

$$Mx - Ny = 2xy\cos xy$$

$$I.F = \frac{1}{Mx - Ny} = \frac{1}{2xy\cos xy}.$$

eqn (1) $\times$ IF

$$\left[\frac{xy^2\sin xy + y\cos xy}{2xy\cos xy}\right]dx + \left[\frac{x^2y\sin xy - x\cos xy}{2xy\cos xy}\right]dy = 0$$

$$\left[\frac{y\tan xy}{2} + \frac{1}{2x}\right]dx + \left[\frac{x\tan xy}{2} - \frac{1}{2y}\right]dy = 0$$

This is of the form $M_1dx + N_1dy = 0$

$$M_1 = \frac{1}{2}y\tan xy + 1/2x.$$

$$N_1 = \frac{1}{2} x \tan xy - \frac{1}{2y}$$

$$\frac{\partial M_1}{\partial y} = \frac{1}{2}\left[xy\sec^2 xy + \tan xy\right]$$

$$\frac{\partial N_1}{\partial x} = \frac{1}{2}\left[xy\sec^2 xy + \tan xy\right]$$

$$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}.$$

Given is an Exact differential equation.

$$G.S = \int\left[\frac{y}{2}\tan xy + \frac{1}{2x}\right]dx + \int -\frac{1}{2y}\,dy = C.$$

$$= \frac{y}{2}.\log\frac{(\sec xy)}{y} + \frac{1}{2}\log x. - \frac{1}{2}\log y = C$$

$$= \frac{1}{2}\left[\log(\sec xy) + \log(x/y) = C\right.$$

$$= \frac{1}{2}\left[\log.\sec xy\,(x/y)\right] = C. //$$

10. $$\frac{dy}{dx} = \frac{x^2+y^2+1}{2xy}$$

$$2xy\,dy = (x^2+y^2+1)\,dx.$$

$$(x^2+y^2+1)\,dx - 2xy\,dy = 0 \quad -①$$

$$M = x^2+y^2+1$$

$$N = -2xy$$

$$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = -2y$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}.$$

This is not EDE

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$$

$$\rightarrow \frac{2y+2y}{-2xy} = \frac{4y}{-2xy} = \frac{-2}{x}$$

$$I.F = e^{\int -\frac{2}{x} dx} = \frac{1}{x^2}$$

eqn ① $\times \frac{1}{x^2}$

$$\left[\frac{x^2+y^2+1}{x^2}\right]dx - \left[\frac{2xy}{x^2}\right]dy = 0$$

$$\left[1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right]dx - \left[\frac{2y}{x}\right]dy = 0$$

This is of the form $M_1 dx + N_1 dy = 0$

$$\frac{\partial M_1}{\partial y} = \frac{2y}{x^2}$$

$$\frac{\partial N_1}{\partial x} = \frac{2y}{x^2}$$

$$\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}.$$

$$G.S = \int \left[1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right]dx + \int 0\, dy = c.$$

$$= x + y^2 \int x^{-2} + \int x^{-2} = c$$

$$= x + y^2\left(\frac{-1}{x^2}\right) + \left(\frac{-1}{x^2}\right) = c$$

$$= x - \frac{y^2}{x^2} - \frac{1}{x^2} = c$$

$$= x - \frac{1}{x^2}\left[y^2 + 1\right] = c.$$

11) $(xy^3+y)dx + 2(x^2y^2+x+y^4)dy = 0$

Given $(xy^3+y)dx + 2(x^2y^2+x+y^4)dy = 0$ — (1)

$M = xy^3+y$

$N = 2(x^2y^2+x+y^4)$

$\frac{\partial M}{\partial y} = 3xy^2+1$

$\frac{\partial N}{\partial x} = 2(2xy^2+1) = 4xy^2+2$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$.

This is not Exact differential equation.

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$$

$$\Rightarrow \frac{3xy^2+1-4xy^2-2}{2(x^2y^2+x+y^4)} = \frac{-xy^2-1}{2(x^2y^2+x+y^4)}$$

Not a function of 'x' alone.

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{4xy^2+2-3xy^2-1}{y(xy^2+1)} = \frac{xy^2+1}{y(xy^2+1)} = \frac{1}{y}$$

$I.F = e^{\int \frac{1}{y} dy} = e^{\log y} = y$

eqn (1) $\times$ I.F

$(xy^4+y^2)dx + 2(x^2y^3+xy+y^5)dy = 0$

It is of the form $M_1dx + N_1dy = 0$

$M_1 = xy^4+y^2$

$N_1 = 2(x^2y^3+xy+y^5)$

$$\frac{\partial M_1}{\partial y} = 4xy^3 + 2y \quad , \quad \frac{\partial N_1}{\partial x} = 4xy^3 + 2y \cdot$$

$$\therefore \frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$$

$$G.S = \int [xy^4 + y^2]\, dx + \int (0y^5)\, dy = c$$

$$= \frac{x^2y^4}{2} + y^2x + \frac{y^6}{3} = c_1$$

b) $x^4 \frac{dy}{dx} + x^3y + \text{cosec}(xy) = 0$

$$x^4\, dy + (x^3y + \text{cosec}(xy))\, dx = 0$$

$$M = x^3y + \text{cosec}(xy)$$

$$N = x^4$$

$$\frac{\partial M}{\partial y} = x^3 - x\,\text{cosec}(xy).\cot(xy)$$

$$\frac{\partial N}{\partial x} = 4x^3$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}.$$

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

$$\frac{4x^3 - x^3 + x\,\text{cosec}(xy).\cot xy}{x^3y + \text{cosec}\, xy}$$

$$3x^3 + x\,\text{cosec}\, xy$$

$$x($$

13. Find orthogonal trajectories of family of circles $x^2+y^2+2\lambda x+c$ where λ is a parameter.

Given $x^2+y^2+2\lambda x+c=0 \longrightarrow (1)$

diff wr to 'x'

$$2x+2y\cdot\frac{dy}{dx}+2\lambda=0$$

$$x+y\cdot\frac{dy}{dx}+\lambda=0$$

$$\lambda=-\left(x+y\cdot\frac{dy}{dx}\right) \longrightarrow (2)$$

Sub 'λ' in eqn (1)

$$x^2+y^2+2\left[-\left(x+y\cdot\frac{dy}{dx}\right)\right]x+c=0$$

$$x^2+y^2-2x^2-2xy\cdot\frac{dy}{dx}+c=0 \longrightarrow (3)$$

To find for O.T

Replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$

(3) $\Rightarrow x^2+y^2-2x^2+2xy\cdot\frac{dx}{dy}+c=0$

$$2xy\cdot\frac{dx}{dy}=-[-x^2+y^2+c]$$

$$\frac{dx}{dy}=-\frac{[-x^2+y^2+c]}{2xy}$$

$$2xy\,dx+(-x^2+y^2+c)\,dy=0$$

~~$y^2-x^2+c=$~~

This is in the form $mx+ny=0$

$$\frac{\partial M}{\partial y}=2x \quad , \frac{\partial N}{\partial x}=-2x. \Rightarrow \quad \therefore \frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}.$$

$$g(y)=\frac{\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}}{M}=\frac{-2x-2x}{2xy}=\frac{-4x}{2xy}=\frac{-2}{y}$$

$\int 2xy\,dx+\int(-x^2+y^2+c)\,dy=0$

$x^2y+-x^2y+\frac{y^3}{3}+cy=0$

$I.F = e^{-\int \frac{2}{y} dy} = e^{-2\log y} = \frac{1}{y^2}$ $\Rightarrow$ Solution

multiply D.E of with I.F

$\frac{1}{y^2}[2xy\,dx + (y^2 - x^2 + c)\,dy] = 0$

$\frac{2x}{y}dx + \left(1 + \frac{x^2}{y^2} + \frac{c}{y^2}\right)dy = 0$

This is of the form $M_1 dx + N_1 dy = 0$

$M_1 = \frac{2x}{y}$, $N_1 = 1 - \frac{x^2}{y^2} + \frac{c}{y^2}$

$\frac{dM_1}{dy} = -2x/y^2$, $\frac{dN_1}{dx} = \frac{-2x}{y^2}$

$\frac{dM_1}{dy} = \frac{dN_1}{dx}$.

$\int \frac{2x}{y}dx + \int \left(1 + \frac{c}{y^2}\right)dy = a$

$\frac{x^2}{y} + y + c\left(-\frac{1}{y}\right) = a$

$x^2 + y^2 - c = ay$

$x^2 + y^2 - ay - c = 0$ //

14) Show that the following curves are self orthogonal curves.

(i) $y^2 = 4a(x+a)$

Given $y^2 = 4a(x+a)$

$y^2 = 4ax + 4a^2 \longrightarrow (1)$

Diff w.r to 'x' in eqn (1)

$2y\frac{dy}{dx} = 4a + 0$

$a = y/2\, dy/dx$.

sub 'a' value in eqn (1)

$y^2 = 4x\left(\frac{y}{2}\cdot\frac{dy}{dx}\right) + 4\left(\frac{y}{2}\cdot\frac{dy}{dx}\right)^2$

$y^2 = 2xy\frac{dy}{dx} + \left(y\frac{dy}{dx}\right)^2 \longrightarrow (2)$

(2) represents D.E of family of curves.

To find D.E of O.T

Replace $\frac{dy}{dx}$ with $\frac{-dx}{dy}$

$y^2 = 2xy\left(\frac{-dx}{dy}\right) + y^2\left(\frac{dx}{dy}\right)^2$

$y^2 = y^2\left(\frac{1}{y^1}\right)^2 - 2xy\left(\frac{1}{y^1}\right)$

$$y^2 = \frac{y^2}{(y_1)^2} - \frac{2xy}{y_1}$$

$$y^2 y_1^2 = y^2 - 2xyy_1$$

$$y^2 = y^2 y_1^2 + 2xyy_1 = 0 \rightarrow (3)$$

(2) = (3)

Hence D.E of family of curves is same as D.E of O.T

∴ Given eqn is self O.T.

(ii) $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1$, λ is a parameter.

Given $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1$ —— (1)

diff w.r to 'x'

$$\frac{2x}{a^2+\lambda} + \frac{2y}{b^2+\lambda}\frac{dy}{dx} = 0$$

$$\frac{x}{a^2+\lambda} + \frac{y}{b^2+\lambda}\frac{dy}{dx} = 0$$

$$x(b^2+\lambda) + y(a^2+\lambda)\frac{dy}{dx} = 0$$

let $\frac{dy}{dx} = p$

$$x(b^2+\lambda) + y(a^2+\lambda)p = 0$$

$$xb^2 + x\lambda + ya^2p + y\lambda p = 0$$

$$\lambda(x+yp) + (xb^2 + ya^2p) = 0$$

$$\lambda = -\frac{(xb^2+ya^2p)}{(x+yp)}$$

$$\Rightarrow a^2+\lambda = a^2 - \frac{(xb^2+ya^2p)}{(x+yp)}$$

$$= \frac{a^2x + ya^2p - xb^2 - ya^2p}{(x+yp)} \quad \therefore = \frac{x(a^2-b^2)}{(x+yp)} \text{ —— (2)}$$

$$b^2+\lambda = b^2 - \frac{(xb^2+ya^2p)}{(x+yp)}$$

$$= \frac{b^2x + yb^2p - xb^2 - ya^2p}{(x+yp)} = \frac{yp(b^2-a^2)}{(x+yp)} \longrightarrow (3)$$

Sub (2) & (3) in eqn (1)

$$\frac{x^2(x+yp)}{x(a^2-b^2)} + \frac{y^2(x+yp)}{yp(b^2-a^2)} = 1$$

$$\frac{x(x+yp)}{(a^2-b^2)} + \frac{y(x+yp)}{p(b^2-a^2)} = 1$$

$$\frac{1}{(a^2-b^2)}(x+yp)(x-y/p) = 1$$

$$(x+yp)(x-y/p) = (a^2-b^2) \longrightarrow (4)$$

(4) represents D.E of family of curves.

To find DE of O.T

Replace $\frac{dy}{dx}$ with $\frac{-dx}{dy}$

Here $p = \frac{dy}{dx}$, $\frac{-dx}{dy} = \frac{-1}{p}$

$$\Rightarrow (x-y/p)(x+yp) = (a^2-b^2) \longrightarrow (5)$$

(4) = (5)

Hence D.E of family of curves is same as D.E of O.T

∴ Given eqn is self O.T.

15) Find the orthogonal trajectories of the following curves:

(i) $r = a(1-\cos\theta)$, a is a parameter.

(ii) $r = 2a(\cos\theta + \sin\theta)$, a is a parameter.

(i) Given $r = a(1-\cos\theta)$ — ①

diff w.r to 'θ'

$$\frac{dr}{d\theta} = 0 + a\sin\theta$$

$$a = \frac{dr}{d\theta}\cdot \operatorname{cosec}\theta$$

Sub in eqn ①

$$r = \frac{dr}{d\theta}\cdot \operatorname{cosec}\theta\,(1-\cos\theta)$$

$$r = \frac{dr}{d\theta}\operatorname{cosec}\theta - \frac{dr}{d\theta}\cot\theta$$

$$r = \frac{dr}{d\theta}\left[\operatorname{cosec}\theta - \cot\theta\right] \text{ — ②}$$

This is DE of family of curves.

For O.T, replace $\frac{dr}{d\theta}$ with $-r^2\frac{d\theta}{dr}$

$$r = -r^2\frac{d\theta}{dr}\left[\operatorname{cosec}\theta - \cot\theta\right]$$

$$\frac{1}{r}dr + \left[\operatorname{cosec}\theta - \cot\theta\right]d\theta = 0$$

$$\int\frac{1}{r}dr + \int\frac{1-\cos\theta}{\sin\theta}d\theta = c$$

$$-\int\frac{1}{r}dr = \int\frac{1-\cos\theta}{\sin\theta}d\theta$$

$$-\log r = \int\frac{2\sin^2\theta/2}{2\sin\theta/2\cdot\cos\theta/2}d\theta + \log c$$

$$-\log r = \int\tan\theta/2\,d\theta + \log c.$$

$$-\log r = 2\log|\sec\theta/2| + \log c.$$

$$-\log r = \log c\cdot(\sec^2\theta/2)$$

$$-\log r = \log c\,(\cos^2\theta/2)$$

$r = c\cdot\cos^2\theta/2$ which is called Eqn of O.T.

(ii) Given $r = 2a(\cos\theta + \sin\theta)$ — (1)

diff w.r.t 'θ'

$$\frac{dr}{d\theta} = 2a(-\sin\theta + \cos\theta)$$

$$2a = \frac{\frac{dr}{d\theta}}{(-\sin\theta + \cos\theta)}$$

sub in eqn (1)

$$r = \frac{\frac{dr}{d\theta}}{-\sin\theta + \cos\theta}(\cos\theta + \sin\theta)$$

$$\frac{r(\cos\theta - \sin\theta)}{\cos\theta + \sin\theta} = \frac{dr}{d\theta}$$

This D.E of family of curves

Replace $\frac{dr}{d\theta}$ with $-r^2\frac{d\theta}{dr}$

$$\frac{r(\cos\theta - \sin\theta)}{(\cos\theta + \sin\theta)} = -r^2\frac{d\theta}{dr}$$

$$\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta} = -r\frac{d\theta}{dr}$$

$$\frac{dr}{r} = -\frac{(\cos\theta + \sin\theta)}{(\cos\theta - \sin\theta)}d\theta$$

$$\int\frac{dr}{r} = \int\frac{-(\cos\theta + \sin\theta)}{(\cos\theta - \sin\theta)}d\theta.$$

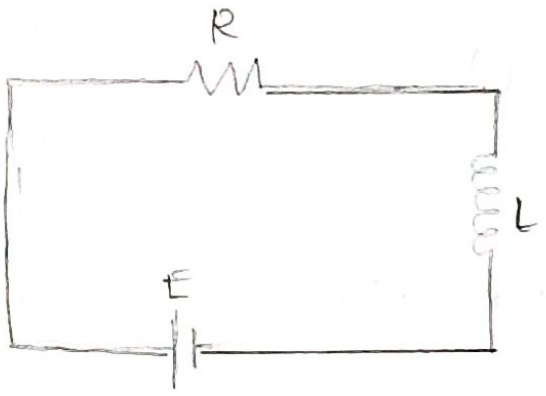
$$\log r = \log(\cos\theta - \sin\theta) + \log c.$$

$$\log r = \log[c(\cos\theta - \sin\theta)]$$

$$r = c.\cos\theta - \sin\theta$$

16) Model R-L circuit, write its differential equation and find the current i at any time t in the circuit.

Consider a circuit with resistance R, inductance L connected in series with emf E, which is called L-R circuit



The voltage drop across R is Ri

The voltage drop across L is $L\frac{di}{dt}$

By Kirchoff's law $Ri + L\frac{di}{dt} = E$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

This is a linear D.E in 'i' with $P(t) = \frac{R}{L}$, $Q(t) = \frac{E}{L}$.

$$I.F = e^{\int R/L\,dt} = e^{\frac{Rt}{L}}$$

$$\text{Gs} \Rightarrow i \cdot e^{Rt/L} = \int \frac{E}{L} \cdot e^{Rt/L}\,dt + c.$$

$$= \frac{E}{L} \cdot \frac{e^{Rt/L}}{(R/L)} + c$$

$$ie^{\frac{Rt}{L}} = \frac{E}{R} \cdot e^{Rt/L} + c.$$

$$i = \frac{E}{R} + c \cdot e^{-Rt/L} \quad \longleftarrow (1)$$

If intially there is no current i.e $i=0$ at $t=0$

$$0 = \frac{E}{R} + ce^{\frac{-R(0)}{L}}$$

$$c = -\frac{E}{R}.$$

$$(1) \Rightarrow i = \frac{E}{R} + \left(\frac{-E}{R}\right)e^{-Rt/L} \Rightarrow \frac{E}{R}\left(1 - e^{-Rt/L}\right),$$

Find the current in electrical circuit given by $L\frac{di}{dt} + Ri = E$, $i(0) = 0$ where $E = 6$ volts, $R = 100$ ohms, $L = 0.1$ henry. Also find how long will be it before the current has reached one half of its maximum value.

⇒ Given $E = 6$ volts, $R = 100$ ohms, $L = 0.1$ henries

$$L\frac{di}{dt} + Ri = E$$

$$\Rightarrow 0.1\frac{di}{dt} + 100i = 6$$

$$\frac{di}{dt} + 1000i = 60$$

This is linear eqn in i with $p(t) = 1000$, $Q(t) = 60$

$$I.F = e^{\int 1000\,dt} = e^{1000t}$$

G.S $\quad i\,(e^{1000t}) = 60\int e^{1000t}\,dt + C$

$$i \cdot e^{1000t} = 60 \cdot \frac{e^{1000t}}{1000} + C$$

$$i \cdot e^{1000t} = \frac{6}{100} \cdot e^{1000t} + C$$

$$i = \frac{6}{100} + C \cdot e^{-1000t} \quad \text{— (1)}$$

when $t = 0$, $i = 0$ gives

$$0 = \frac{6}{100} + C e^{0}$$

$$C = -\frac{6}{100}$$

(1) ⇒ $\quad i = \frac{6}{100} - \frac{6}{100} e^{-1000t}$

$$i = \frac{6}{100}\left[1 - e^{-1000t}\right] \quad \text{— (2)}$$

Also

Given that $i = \frac{1}{2}$ [max value of i]

from (2) the max value of i is $\frac{6}{100}$

$$i = \frac{1}{2}\text{ [max value of i]} \Rightarrow \frac{6}{100}\left[1 - e^{-1000t}\right] = \frac{1}{2}\left[\frac{6}{100}\right]$$

$$\Rightarrow 1 - e^{-1000t} = \frac{1}{2}$$

$$e^{-1000t} = \frac{1}{2}$$

$$\log\left[\frac{1}{2}\right] = -1000t$$

$$\log\left[2^{-1}\right] = -1000t$$

$$-\log[2] = -1000t$$

$$\frac{\log[2]}{1000} = t$$

$$t \simeq 0.000301,$$